ROS1 fusion-hotspot benchmark report

Study: luad_tcga_pan_can_atlas_2018

Abstract

This report analyzes ROS1 gene fusions from the luad_tcga_pan_can_atlas_2018 study, covering 10 fusion events. 8/8 fusions with known frame status (100.0%) were in-frame. 8/10 fusions (80.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). No literature benchmark is configured for ROS1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 10 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was CD74, observed in 3 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 8/8 (100.0%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.152847.

Domain disruption

Disruption of the Fibronectin type III domain was tested with Fisher's exact test comparing in-frame fusions against all others; 7/7 (100.0%) of in-frame fusions disrupted the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.001998.

Junction concentration

71% of 7 fusions with this gene as the 3' partner join at exon 34, against 1% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection could not determine a breakpoint boundary for this run: all events share a single outcome class; no separation is possible

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=7) and "not_retained" (n=0). For Frame_status, retained: 7/7 (100.0%, 95% CI 64.6%-100.0%); not_retained: 0/0 (unavailable, 95% CI unavailable-unavailable). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: ROS1: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (3 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: ROS1 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: ROS1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: all events share a single outcome class; no separation is possible

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CD74	3	0.3	0.08157426466617199	0.26994040818153375	None	0.6456695649333126	confidence_certainty, domain_disruption, domain_retention, recurrence	0.2693512991719777	1
CLTC	2	0.2	0.08157426466617199	0.26994040818153375	None	0.6456695649333126	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23405718152491883	2
EZR	2	0.2	0.08157426466617199	0.26994040818153375	None	0.6456695649333126	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23405718152491883	3
SLC34A2	2	0.2	0.08157426466617199	0.26994040818153375	None	0.6456695649333126	confidence_certainty, domain_disruption, domain_retention, recurrence	0.23405718152491883	4
FBXO9	1	0.1	0.08157426466617199	0.26994040818153375	None	0.6456695649333126	confidence_certainty, domain_disruption, domain_retention, recurrence	0.19876306387786002	5

domain_disruption tables

frame_domain_contingency_table

7	0
0	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
LUAD	[7, 0], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
fibronectin	Fibronectin type III domain	7	0	0	7	0.0

domain_retention tables

frame_domain_contingency_table

7	0
0	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
LUAD	[7, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	7	7	0	0	None

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
36	7	7	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CD74	3	3	2	2
CLTC	2	2	1	1
EZR	2	2	2	2
FBXO9	1	1	1	1
SLC34A2	2	2	2	2

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
6	117324000	117325000	5	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5
6	117321000	117322000	1	EVT-TCGA-44-2665-01-3
6	117329000	117330000	1	EVT-TCGA-62-A46Y-01-10

exact_position_recurrence

chromosome	position	n_events	event_ids
6	117324415	5	EVT-TCGA-05-4426-01-9, EVT-TCGA-55-6986-01-4, EVT-TCGA-64-1680-01-1, EVT-TCGA-86-8278-01-2, EVT-TCGA-NJ-A7XG-01-5

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
1853	5	1	0	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	34	5	7	0.7142857142857143	0.013754398208729536	7.995584330671725
three_prime	32	1	7	0.14285714285714285	0.05701677862107069	0.4723809411404404
three_prime	35	1	7	0.14285714285714285	0.022339992439442846	0.8348121040437931

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-64-1680-01-1	TCGA-64-1680-01	TCGA-64-1680	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	1173244145	34	1853	False	1853	True	exact match	retained	1.0	False													Protein kinase domain	Fibronectin type III domain		NA | CD74-ROS1 fusion	in-frame
EVT-TCGA-86-8278-01-2	TCGA-86-8278-01	TCGA-86-8278	Non-Small Cell Lung Cancer	LUAD	CD74-ROS1	CD74	in-frame	three_prime	1173244145	34	1853	False	1853	True	exact match	retained	1.0	False													Protein kinase domain	Fibronectin type III domain		NA | CD74-ROS1 fusion	in-frame
EVT-TCGA-44-2665-01-3	TCGA-44-2665-01	TCGA-44-2665	Non-Small Cell Lung Cancer	LUAD	CLTC-ROS1	CLTC	in-frame	three_prime	1173213944	35	1881	False	1881	True	exact match	retained	1.0	False													Protein kinase domain	Fibronectin type III domain		NA | CLTC-ROS1 fusion	in-frame
EVT-TCGA-55-6986-01-4	TCGA-55-6986-01	TCGA-55-6986	Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	1173244145	34	1853	False	1853	True	exact match	retained	1.0	False													Protein kinase domain	Fibronectin type III domain		NA | EZR-ROS1 fusion	in-frame
EVT-TCGA-NJ-A7XG-01-5	TCGA-NJ-A7XG-01	TCGA-NJ-A7XG	Non-Small Cell Lung Cancer	LUAD	EZR-ROS1	EZR	in-frame	three_prime	1173244145	34	1853	False	1853	True	exact match	retained	1.0	False													Protein kinase domain	Fibronectin type III domain		NA | EZR-ROS1 fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-86-8278-01-6	TCGA-86-8278-01	TCGA-86-8278	Non-Small Cell Lung Cancer	LUAD	ROS1-CD74	CD74	unknown	five_prime	117379059	17	866	False	865	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Protein kinase domain; Fibronectin type III domain		NA | ROS1-CD74 fusion	frame shift

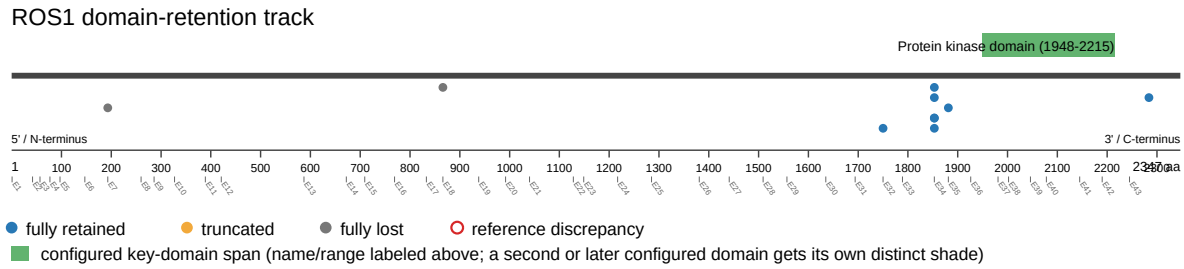
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-44-2665-01-7	TCGA-44-2665-01	TCGA-44-2665	Non-Small Cell Lung Cancer	LUAD	ROS1-CLTC	CLTC	in-frame	five_prime	117288686	43	2284	False	2283	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False															Protein kinase domain; Fibronectin type III domain			NA | ROS1-CLTC fusion	in-frame

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-64-1680-01-8	TCGA-64-1680-01	TCGA-64-1680	Non-Small Cell Lung Cancer	LUAD	ROS1-FBXO9	FBXO9	unknown	five_prime	117403139	6	193	False	192	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Protein kinase domain; Fibronectin type III domain		NA | ROS1-FBXO9 fusion	frameshift
EVT-TCGA-05-4426-01-9	TCGA-05-4426-01	TCGA-05-4426	Non-Small Cell Lung Cancer	LUAD	SLC34A2-ROS1	SLC34A2	in-frame	three_prime	117324415	34	1853	False	1853	True	exact match	retained	1.0	False												Protein kinase domain	Fibronectin type III domain		NA | SLC34A2-ROS1 fusion	in-frame

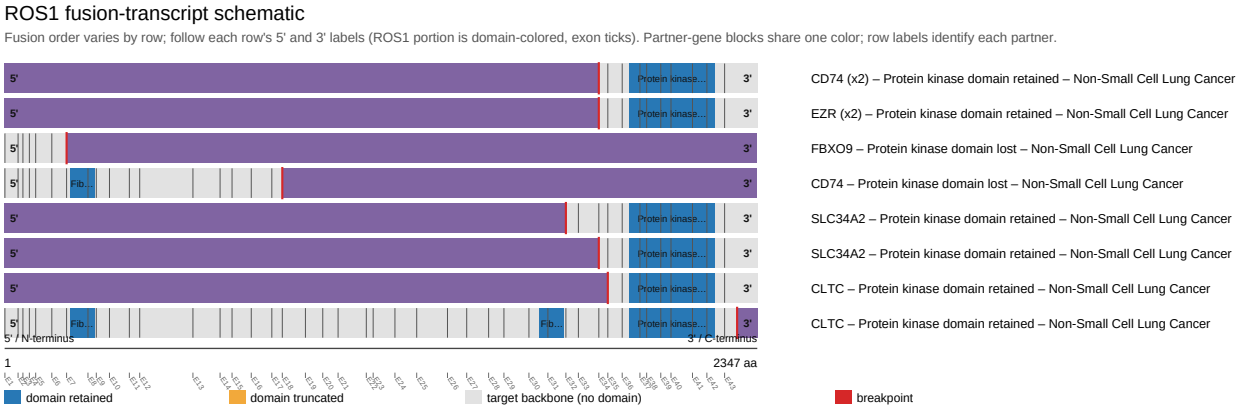
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-62-A46Y-01-10	TCGA-62-A46Y-01	TCGA-62-A46Y	Non-Small Cell Lung Cancer	LUAD	SLC34A2-ROS1	SLC34A2	in-frame	three_prime	117329446	32	1750	False	1750	True	exact match	retained	1.0	False												Protein kinase domain	Fibronectin type III domain		NA | SLC34A2-ROS1 fusion	in-frame

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.